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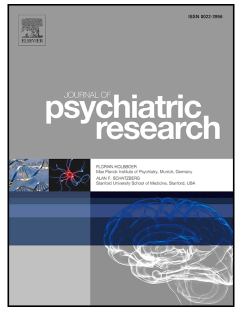

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Analysis on speech Signal Features of manic patients

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ABSTRACT

Given the lack of effective biological markers for early diagnosis of bipolar mania, and the tendency for voice fluctuation during transition between mood states, this study aimed to investigate the speech features of manic patients to identify a potential set of biomarkers for diagnosis of bipolar mania. 30 manic patients and 30 healthy controls were recruited and their corresponding speech features were collected during natural dialogue using the Automatic Voice Collecting System. Bech-Rafaelsdn Mania Rating Scale (BRMS) and Clinical impression rating scale (CGI) were used to assess illness. The speech features were compared between two groups: mood group (mania vs remission) and bipolar group (manic patients vs healthy individuals). We found that the characteristic speech signals differed between mood groups and bipolar groups. The fourth formant (F4) and Linear Prediction Coefficient (LPC) ($P < 0.05$) were significantly differed when patients transmitted from manic to remission state. The first formant (F1), the second formant (F2), and LPC ($P < 0.05$) also played key roles in distinguishing between patients and healthy individuals. In addition, there was a significantly correlation between LPC and BRMS, indicating that LPC may play an important role in diagnosis of bipolar mania. In this study we traced speech features of bipolar mania during natural dialogue (conversation), which is an accessible approach in clinic practice. Such specific indicators may respectively serve as promising biomarkers for benefiting the diagnosis and clinical therapeutic evaluation of bipolar mania.

Key Words: Bipolar mania; speech signal features; objective biological markers.

1. INTRODUCTION

Bipolar disorder (BP) is one of the top ten most disabling disorders in working age adults (Young AH., et al., 2011; JJ, Chen et al., 2014), with the 2.4% lifetime prevalence according to the World Mental Health Survey Initiative (Merikangas KR, et al., 2011). It's responsible for the loss of more disability-adjusted life years than all forms of cancer or major neurologic conditions such as epilepsy and Alzheimer disease (JJ Guilbert, 2002). Individuals with BP experience high rates of suicide (Angst et al., 2002; Alloy LB, 2015), with 20–30 times greater risk than the general population (Pompili, et al., 2013). The economic impact is significant as well; each patient with BP incurs lifetime costs of up to $600,000 in the USA (Begley CE, et al., 2001). Thus, BP carries a profound burden of illness and disability, which has a significant, negative impact on sufferers' lives.

Mania is one of the stages of BP, characterized by a pathologically elevated mood. Patients with mania are also more likely to have diminished ability to control impulses, and to have more difficulties avoiding destructive and threatening behavior, and therefore make harmful choices in their daily-life and professional functions. In general, the methods for diagnosing mania have relied purely on self-report and clinician judgments of symptom severity (Frey BN, et al., 2013), such as the Young Mania Rating Scale (YMRS) (Young RC, 1978), Bech-Rafaelsen Mania Rating Scale (BRMS) (Bech and Rafaelsen, 1978), Mini-international neuropsychiatric interview (MINI) (Sheehan D, et al., 1998), Mood Disorder Questionnaire (MDQ) (Hirschfeld, et al., 2000, 2003b). However, clinical methods may be biased and depend on a very experienced clinician to interpret optimally and may be difficult to arrive at a diagnosis. It is urgent and necessary to identify objective measures for early prediction and identification of mood states as well as measures of treatment efficacy.

The emotional information in the speech signal is an important source of information. Studies of emotional speech have shown that emotion change can be associated with changes in the prosodic and spectral characteristics of speech signals (Bulut et al., 2005, 2002; Yamauchi, H. et al., 2000); the same word, because of the variation in speaker's emotion, may have even greater variation in their speech features. Previous speech studies in depression suggest that acoustic features are useful biomarkers for evaluation of symptom severity and treatment efficacy for depression (Stassen HH, et al., 1998; Alpert M, et al., 2001; Mundt JC, et al., 2007, 2012; II EM, et al., 2008; Low LSA, et al., 2011; Ooi KEB, et al., 2013). Bipolar disorder has significant changes in emotion and acoustic features between different states (e.g., euthymia, depressive and manic states) (Gideon J, et al., 2016). Thus, we hypothesize that changing speech signal features could be used to recognize the onset of a manic episode.

The collection and extraction of speech features is a key step in speech recognition (Esposito A, et al., 2012; P. Liu, et al., 2012). A method of speech sample collection in a natural condition is very important for assessment of acoustic features. To date, the common methods of collecting speech samples have been structured, primarily reading a specified article, which is more stylized and mechanized, and may introduce a deviation and artificial nature to the true quality of speech.

In the current study, we collected the speech samples of patients with mania and healthy controls under a natural dialogue state, removed background noise, which help to accurately assess the relationship between speech features and illness.

2. MATERIALS AND METHODS

2.1. Subjects

Thirty manic patients (14 males) and 30 healthy controls (16 males) between 18 and 65 years old were recruited. All subjects spoke fluent native Mandarin Chinese. The patients met the diagnostic criteria of bipolar I disorder with the Structured Clinical Interview Mimi-International Neuropsychiatric Interview (MINI) for DSM-IV. The severity of symptoms reached a BMRS score of 13 or greater at baseline, the onset of the current episode was within the past year. Manic patients who suffered from severe physical disease, substance abuse, harmful and impulsive suicide attempts, received modified electric convulsive therapy (MECT) within the past month, or were treated with medication considered to affect language function were excluded.

Controls were recruited from the community, and manic patients from in-patient clinics of Shanghai Mental Health Center, Shanghai, China, between October 2014 and June 2015. All participants completed written informed consent, and agreed to use smart phone to record the voice.

30 manic patients completed the study at baseline, 25 manic patients completed follow-up, and in remission state re-assessed with BRMS and CGI, meanwhile recorded speech samples.

2.2. Speech sample collection and clinical assessment.

Android mobile phones equipped with an automatic voice collecting system which developed by us were used to collecting speech samples. The software not only could record samples in a single-channel, but also could upload collected voice signals to dedicated server for sorting and analyzing data in real-time.

Participants were placed in a quiet indoor room where the surrounding noise was controlled under 30 db. They assumed a natural and comfortable sitting posture, holding the smart phone installed with the voice collecting system software, and had a telephone call with a psychiatrist. The call duration was 25 min -30 min each time. All speech samples were saved as single channel, aac files, sampled at 8KHz, 8-bit. The control group attended one time visit to collect the speech sample. The manic patients were interviewed twice during the hospitalization. The baseline data were collected in the first clinical interview in a manic state and the follow-up data collected when they returned to remission state with BRMS score of 6 or lower. All the speech samples were free (conversational) speech. After the call, the psychiatrist assessed the clinical symptoms of the patients.

BRMS and Clinical Global Impression **(**CGI) were used to assess the severity of the patients' symptoms. BRMS consists of 11 items and each item has five-grade scores (0-4). The main outcome of our study was assessment of speech features and the correlation between BRMS score and speech features. CGI was used to assess the severity of mania and global improvement. The patient is rated from one (normal, not ill) to seven (Very severely ill) on the interview. The assessment time was with BRMS simultaneously.

2. 3. Speech features extraction and statistical analysis

Speech signal features were extracted and analyzed by our cooperative team from

Department of Electronic Engineering, Shanghai Jiao Tong University, Shanghai, China. After samples training and speech signal processing, some vocal acoustic features with higher recognition were extracted (Xu D, et al., 2013; Gui C, et al., 2015). The features consisted of the first formant to sixth formant (F1, F2, F3, F4, F5, F6; unit: Hz), Formant Bandwidth (unit: Hz), Formant Amplitude (unit: dB), Linear Prediction Coefficient (LPC), and Mel Frequency Cepstrum Coefficient (MFCC). Then we analyzed data using the Statistical Analysis System of the Social Science (SPSS) version 17.0 (SPSS Inc., Chicago, IL, USA.). A two-sided value of $p < 0.05$ was specified as statistical significance. Chi-square ($\chi 2$) tests, t-tests and Mann-whitney U-tests were used to compare the differences inter-group and between-groups (eg: the manic state versus the remission state; the manic group versus the control group). Pearson's correlation analysis was used to test for the correlation between BRMS and speech features.

3. RESULTS

3.1. Demographic and clinical features.

Table 1 summarizes the demographic data of subjects in both groups. In our study, 30 patients with definite diagnosis of bipolar mania and 30 controls were involved. The manic subjects included 14 males and 16 females with a mean (SD) age of 41.40 (11.44) years and a mean (SD) onset age of 26.89 (9.23) years, their mean duration of illness was 15.26 (10.82) years. The control subjects included 16 males and 14 females with a mean (SD) age of 36.29 (13.73) years. There was no significant difference between groups among their age, sex, educational level ($p > 0.05$). Table 2 shows the mean (SD) scores of BRMS and CGI-BP-S were 26.34 (8.56) and 5.37 (0.88) at baseline, respectively. When the study ended, the mean (SD) scores of BRMS, CGI-I and CGI-BP-S were 2.2 (1.57), 1.04 (0.20) and 1.07 (0.26).These changes were statistically significant ($p< 0.001$).

3.2. Inter-Groups Analysis on voice features of speech.

To investigate the distinction between manic and healthy control subjects, we compared these ten voice features of speech by the qualitative analysis (table 3**.** And we found that F1 (p= 0.000), F2 (p = 0.036), LPC (p = 0.000) were significantly higher in manic group than that of in controls. No other contrasts were significant (table 3). Moreover, Figure 1 intuitively identified the distribution of the differences in F1, F2 and LPC between manic group and controls.

Pearson's correlation analysis showed that there was a significantly correlation between LPC and the scores of BRMS (p = 0.040), but no significant association between BRMS and F1, F2 (table 4).

3.3. Inner-Group Analysis on voice features of speech.

As shown in Table 5 and Figure 2, we did an inner-group analysis on voice features of speech to compare sufferers who were in manic state with those who were in remission state. We found that F4 (t = 3.570, p = 0.003) and LPC (t = 4.083, p = 0.001) were significantly higher in manic state than those in remission state.

4. DISCUSSION

In this study, we investigated the distinct variations from speech signal features aimed at identifying objective biomarkers for Bipolar Mania by comparing measures in the manic state with remission state in manic patients and comparing manic patients with healthy people. Several of our findings are consistent with previous studies of speech signal features in bipolar disorder, and the effectiveness of using smart-phone to collect speech samples (Osmani V, et al., 2013; Grunerbl et al., 2012, 2014; Xu c et al., 2013; Gideon J et al., 2016; Hung GC, et al., 2016), and identify significant variations in voice features of speech with mood states (Muaremi A, et al., 2014; Guidi A, et al., 2015).

Osmani V, et al., Grunerbl et al., Xu c et al. (Osmani V, et al., 2013; Grunerbl et al., 2012, 2014; Xu c et al., 2013) studied bipolar episodes and found positive effects of using smart-phone for tracing mobility and speech cues. Variations in speech signal features of bipolar disorder have also been reported in studies such as Vanello N et al., Karam ZN et al., Muaremi A et al., and Guidi A et al (Vanello N et al., 2012; Karam ZN et al., 2014; Muaremi A, et al., 2014; Guidi A, et al., 2015). Using neutral text reading and TAT (Thematic Apperception Test) images elicitation to collect speech samples for speech analysis in 6 bipolar patients, results from Vanello N et al. (Vanello N, et al., 2012) showed that the changes in average pitch, jitter, and pitch standard deviation were found in bipolar patients. Muaremi A et al. (Muaremi A, et al., 2014) extracted three different types of features, namely phone call statistics, social signals derived from the phone call conversation and acoustic emotional properties of the voice in 12 bipolar patients, then they used the random forest classifier (RF) to train and test person-dependent models. By fusing all features together, they were able to predict the bipolar states with an average F1 score of 82%. Guidi A et al. (Guidi A, et al., 2015) presented an automatic method to characterize fundamental frequency (F0) dynamics in voiced part of syllables in 10 bipolar patients and highlighted significant changes of the features with different mood states. Several reports focused on designing new classifiers which could be used to classify the patient's mental health. Karam ZN et al. (Karam ZN, et al., 2014) indicated that hypomania and depression could be differentiated from euthymia by using speech-based classifiers such as LiblinearSVM and libSVM (Support Vector Machine, SVM). In 2016, Hung GC et al. (Hung GC, et al., 2016) provided preliminary evidence to predict concurrent negative emotions via mobile phone usage patterns with substantial accuracy that by adapting sophisticated machine-learning methods. Additional support was provided by the results of studies by Gideon J et al. (Gideon J, et al., 2016), they investigated into acoustic variations caused by recording with two types of phones, and demonstrated that through some pipeline modifications include certain preprocessing, feature extraction, and data modeling techniques it was possible to have a higher performance in statistically significance, which highlights the potential of distributed mental health systems. This approach is supported by our team, demonstrating a higher classification for participant-independent to classify mental states. Though these relatively few studies, the results suggested that in bipolar individuals there are variations in several features of speech, that could considered for diagnosing or predicting mood states.

However, unlike the published scientific literature, our findings also suggest novel and un-researched aspects of speech signal features phenomenology. Specifically, we focus on collecting speech samples under a natural conversational state; analyzing speech features in two mood episodes during hospitalizations to avoid data loss during the follow-up period. We measured the clinical symptoms in the recruited subjects as the same time as every speech signal collection. Besides determining which features are the most relevant, we also fused the features to establish the classification performance. We found that the SVM classifier was adopted as the main tools for classifying the mental state for a single patient, and the Gaussian mixture model (GMM) classifier yielded the best performance with a classification rate of 72% for multi patients (Gui C, et al., 2015). However, discrepancies between our study and other studies mentioned above could be caused by differences in populations and data samples. The previous data samples were very small, only 6, 10 or 12. To improve the evaluation, more subjects are necessary. Thus in our research, we collected 30 data samples and completed 25 speech samples of two episodes, then found not only prosodic features but also spectral features played a great role in bipolar patients.

The vocal features of speech play a crucial role in conveying the speakers' affective disposition and emotional state to listeners. BP is a common relapsing mental disorder with no empirical and objective biomarkers to facilitate diagnosis and efficacy evaluation. During mania episode an individual behaves or feels abnormally energetic, happy or irritable, which leads to variations of speech features. It is expected that features will be distinctive when it comes to acoustic analysis of manic patients. In this report, two types of comparison are worth drawing. The first involves contrasts between manic and control subjects in the speech signal features. The second involves contrasts in the same way that the speech signal features differentiate between manic state and remission state; several significant contrasts emerged. Mania showed higher F1, F2, and LPC than controls and these three features had distinct variation in manic persons.

Therefore, it appears that the effects on the speech characteristics of mania are more profound than the effects on the speech characteristics of control subjects. When comparing the manic episode with remission, manic episode showed higher F4 and LPC. While patients changed from the manic state to remission state, several speech signal features had variations. Hence, it was likely that the changing values in F4 and LPC reflected different mental states of a sufferer, and therefore, these contained important cues for the recognition of clinical state and efficacy evaluation by speech. In addition, we detected a significant and positive correlation between LPC and BRMS score. This suggests that LPC may play an important role in diagnosis of bipolar mania. The abnormally happy or irritable states may increase LPC during mania. When LPC rises beyond a certain threshold, the risk of emerging bipolar mania should be considered. LPC was found to be of considerable significance, making it an effective feature for discrimination bipolar episode either manic or normal, or distinguishing manic patients from healthy individuals. The extent to the variation of LPC in relation to clinical severity and treatment efficacy, and possibly early recognition and warning is also worthy of further study. The clinical manic state may have a significant effect on speech signal dysfunction, which in turn, makes speech effective to determine manic states that are patient dependent.

In the present paper, F1, F2, F4 are prosodic features, and LPC is one of the spectral features. As mentioned earlier, prosodic features have been widely used in speech emotion recognition and demonstrated considerable recognition success, nevertheless, spectral features, based on the short-term power spectrum of sound, have received less attention in emotion recognition. Prosodic features reflect the intensity and intonation of the voice, and make the voice more natural, which have been found great use in a variety studies related to emotion and stress recognition (Ververidis

D, et al., 2004; Fernandez R, et al., 2005). In the research, we sampled conversational speech so as to make the voice more natural, and F1, F2, F4 have thus clearly been shown to be of importance for the recognition of sufferers' emotional states. Moreover, spectral features, which are mainly divided into spectral features and cepstral features, reflect the spectral characteristics of emotion speech. They are another effective group of features for describing emotional states (Banse, R, et al., 1996). Subsequent work has showed that spectral features exhibited the best performance as well, and has been widely used (Wang YT, et al., 2013; Nica A, et al., 2006). However, most of the researches were tested on healthy people. In our study, we found LPC have had better performance for emotion recognition in clinical mania. It is interesting to note that LPC was the speech feature that was best predicted in our study. Hence, this raises the hypothesis that the potentiality of speech signal analysis might be feasible, and adding more multiple features to the speech signal to find more strong predictors as objective biomarkers for bipolar mania might be promising for future study.

This research successfully highlights features combining prosodic information and spectral information for charactering manic speech, however, there were still some limitations, such as the short observational time, and the patients were all accepting drug treatment during the period of observation, although they has little side-effects from drugs. To recruit more first-episode and drug naïve bipolar patients and enlarge the sample size and prolong the observational time on the feature selection strategy are critical important for finding the objective biomarker of bipolar mania in the future research.

The importantance of considering the impact of speech signal features as vocal signs for the prediction and diagnosis of bipolar disorders is emphasized. We anticipate further studies that focus specially on this to form diagnostic paradigms for bipolar patients. Possibly in the near future, the speech signal will upload to the cloud servers automatically as the patients use their smartphones. Applying diagnostic criteria, we posit that delivering patient reports to their own terminals, or those of their guardians and doctors will enable prediction and management and will be soon feasible and worthwhile.

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Figure.1. The distribution of F1, F2 and LPC between manic group and controls

Figure 2. The distribution of the differences of F4 and LPC between manic state and remission state

Table 1. Demographic, clinical features of the subjects:

	Manic subjects	Control subjects	Significance
N	30	30	
Age [a]	41.40 (11.44)	36.29 (13.73)	$p = 0.100$
Sex [b]			$p = 0.267$
Men	14 (46.7%)	16 (53.3%)	
Women	16 (53.3%)	14 (46.7%)	
Education (within-group %) [b]			$p = 0.987$
1	16.7	16.7	
2	20.0	23.3	
3	33.3	33.3	
4	30.0	26.7	
Duration of illness	15.26 (10.82)	-	
Onset age	26.89 (9.23)	-	

[a] t-test

[b] Chi-square

Education 1: Elementary school, 2: junior high school, 3: technical secondary school or high school, 4: university.

Table 2. **Baseline and end line scores of BRMS, CGI-I, CGI-BP-S**

	N	BRMS	CGI-I	CGI-BP-S
Baseline	30	26.34 ± 8.56	-	5.37 ± 0.88
End line	25	2.2 ± 1.57	1.04 ± 0.20	1.07 ± 0.26
P	-	< 0.001 **	< 0.001 **	< 0.001 **

** $p < 0.001$.

Table 3. **The differences in voice features of speech between-groups**

voice features	manics (n=30)	controls (n=30)	*t*/U test	*p*
F1 [a]	0.046 (0.010)	0.041 (0.006)	-3.699	< 0.001**
F2 [b]	0.089 (0.010)	0.083 (0.009)	2.153	0.036*
F3 [a]	0.180 (0.019)	0.182 (0.014)	-0.161	0.872
F4 [b]	0.264 (0.015)	0.259 (0.012)	1.241	0.221
F5 [a]	0.362 (0.021)	0.358 (0.014)	-1.347	0.178
F6 [b]	0.419 (0.012)	0.426 (0.015)	-1.953	0.058
Bandwidth [b]	19.704 (7.293)	18.840 (5.996)	0.455	0.651
Amplitude [a]	0.042 (0.006)	0.043 (0.003)	-0.523	0.601
LPC (Log) [b]	-0.575 (0.067)	-0.647 (0.067)	3.756	< 0.001**
MFCC [b]	-0.279 (0.189)	-0.183 (0.222)	-1.611	0.115

[a] F1, F3, F5, Amplitude used Mann-whitney U-test, values expressed as median (inter-quartile range).

[b] F2, F4, F6, Bandwidth, LPC and MFCC used independent sample T-test, values expressed as mean (SD).

* $p < 0.05$.

** $p < 0.001$.

Table 4. **Voice features of speech associated with BRMS scores**

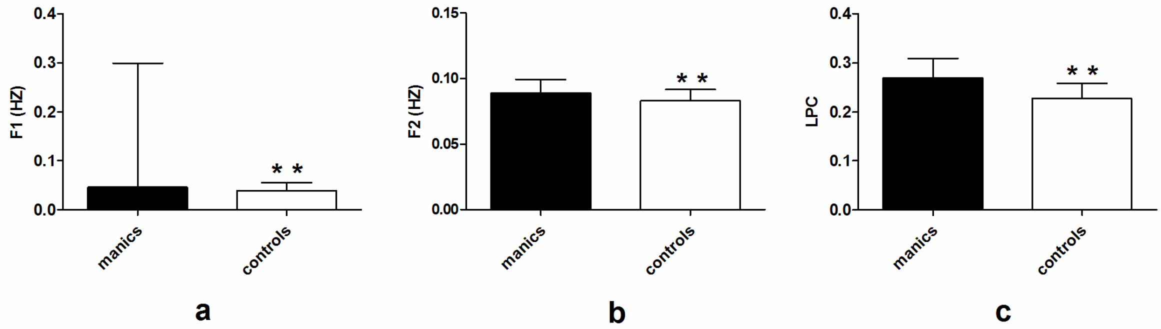
	F1	**F2**	**LPC**
BRMS	-0.201（0.306）	0.263（0.184）	0.398（0.040*）

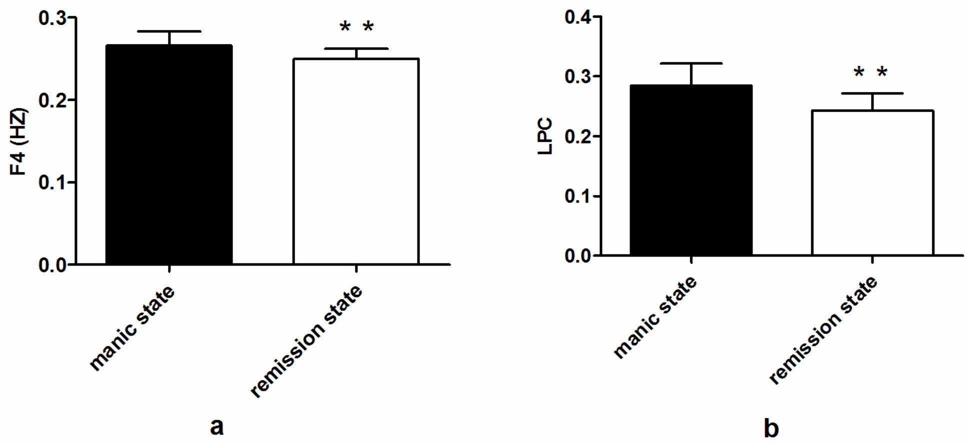
* significant correlation ($p < 0.05$)

Table 5. **The vocal acoustic measures in manic state and remission state**

VA measure	Mania (n=25)	Remission (n=25)	*t* test	*p*
F1	0.046 (0.064)	0.044 (0.006)	0.904	0.381
F2	0.090 (0.011)	0.090 (0.008)	0.123	0.904
F3	0.185 (0.014)	0.182 (0.010)	0.636	0.535
F4	0.266 (0.017)	0.250 (0.013)	3.570	0.003**
F5	0.362 (0.165)	0.359 (0.019)	0.480	0.638
F6	0.416 (0.015)	0.418 (0.012)	-0.331	0.745
Bandwidth	20.562 (6.751)	18.739 (5.003)	0.838	0.416
Amplitude	0.041 (0.005)	0.040 (0.005)	0.580	0.571
LPC	0.284 (0.037)	0.243 (0.029)	4.083	0.001**
MFCC	-0.340 (0.151)	-0.337 (0.174)	-0.121	0.906

** $p < 0.01$


0.4
0.3
0.2
0.1
0.0
F1 (HZ)
* *
manics
controls
a
0.15
0.10
0.05
0.00
F2 (HZ)
* *
manics
controls
b
0.4
0.3
0.2
0.1
0.0
LPC
* *
manics
controls
c


0.3
0.2
0.1
0.0
F4 (HZ)
* *
manic state
remission state
a
0.4
0.3
0.2
0.1
0.0
LPC
* *
manic state
remission state
b

Conflict of interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We confirm that we have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. In so doing we confirm that we have followed the regulations of our institutions concerning intellectual property.

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